



Lesson 5.4 — Transformations of Root Functions

Big idea: Same transformation rules as before: shift, stretch, reflect. The only thing that changes between square and cube roots is the parent shape.

Quick review

General form: $g(x) = a \cdot \sqrt[n]{b(x - h)} + k$.

(h, k): the “anchor” point (origin of the parent shifted to (h, k)).

a: vertical stretch (and reflection if $a < 0$).

b: horizontal compression (and reflection if $b < 0$); usually we leave b inside the radical and treat it as a domain shift trick.

Worked example

Worked example. Describe transformations from $y = \sqrt{x}$ to $y = -2\sqrt{x + 4} - 1$.

Shift left 4, vertical stretch by 2, reflect across x-axis, shift down 1. Starts at $(-4, -1)$ and decreases.

Warm-ups

1. Describe the transformation from $y = \sqrt{x}$ to $y = \sqrt{x - 7}$.
2. Describe the transformation from $y = \sqrt{x}$ to $y = \sqrt{x} + 5$.
3. Describe the transformation from $y = \sqrt{x}$ to $y = -\sqrt{x}$.

Practice

1. Sketch (describe) $y = \sqrt{x + 2} - 3$. Starting point and domain.
2. Sketch (describe) $y = \sqrt{x - 1} - 4$. Inflection point and a couple of points.
3. Write a transformed square-root function whose graph starts at $(3, -2)$ and is reflected across the x-axis.

Challenge

1. If the graph of $f(x) = \sqrt{x}$ is shifted right 5, vertically stretched by 4, and reflected across the y-axis, write the equation $g(x)$. State its domain.
2. Two functions: $f(x) = \sqrt{x - 2}$ and $g(x) = \sqrt{2 - x}$. Describe how their graphs differ and the domain of each.



Answer key — Lesson 5.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Shift right 7.
2. Shift up 5.
3. Reflect across the x-axis.

Practice

1. Starts at $(-2, -3)$. Domain: $x \geq -2$.
2. Inflection $(1, -4)$. Points: $x = 0 \rightarrow \blacksquare(-1) - 4 = -5$. $x = 2 \rightarrow \blacksquare 1 - 4 = -3$. $x = 9 \rightarrow 2 - 4 = -2$.
3. $y = -\sqrt{(x - 3)} - 2$.

Challenge

1. $g(x) = 4\sqrt{-(x - 5)} = 4\sqrt{5 - x}$. Domain: $5 - x \geq 0 \Rightarrow x \leq 5$. (Or: reflected across y-axis means replacing x with $-x$ in the inside.)
2. f starts at $(2, 0)$, goes right and up. g starts at $(2, 0)$, goes LEFT and up (reflected horizontally). Domain f : $x \geq 2$. Domain g : $x \leq 2$. They're mirror images across $x = 2$.